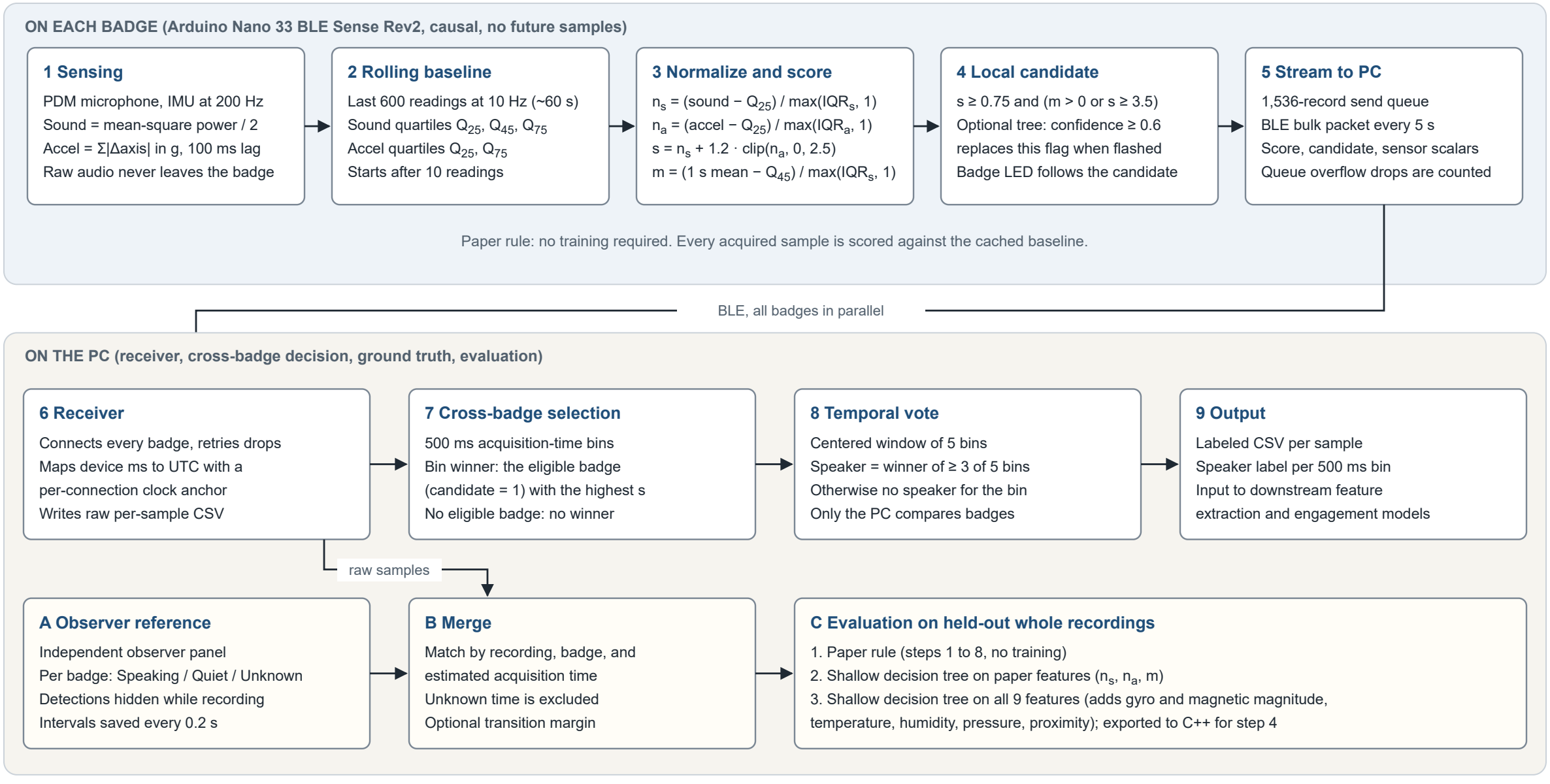

ON EACH BADGE (Arduino Nano 33 BLE Sense Rev2, causal, no future samples)
1 Sensing
PDM microphone, IMU at 200 Hz
Sound = mean-square power / 2
Accel = Σ|Δaxis| in g, 100 ms lag
Raw audio never leaves the badge
2 Rolling baseline
Last 600 readings at 10 Hz (~60 s)
Sound quartiles Q25, Q45, Q75
Accel quartiles Q25, Q75
Starts after 10 readings
3 Normalize and score
ns = (sound − Q25) / max(IQRs, 1)
na = (accel − Q25) / max(IQRa, 1)
s = ns + 1.2 · clip(na, 0, 2.5)
m = (1 s mean − Q45) / max(IQRs, 1)
4 Local candidate
s ≥ 0.75 and (m > 0 or s ≥ 3.5)
Optional tree: confidence ≥ 0.6
replaces this flag when flashed
Badge LED follows the candidate
5 Stream to PC
1,536-record send queue
BLE bulk packet every 5 s
Score, candidate, sensor scalars
Queue overflow drops are counted
Paper rule: no training required. Every acquired sample is scored against the cached baseline.
BLE, all badges in parallel
ON THE PC (receiver, cross-badge decision, ground truth, evaluation)
6 Receiver
Connects every badge, retries drops
Maps device ms to UTC with a
per-connection clock anchor
Writes raw per-sample CSV
7 Cross-badge selection
500 ms acquisition-time bins
Bin winner: the eligible badge
(candidate = 1) with the highest s
No eligible badge: no winner
8 Temporal vote
Centered window of 5 bins
Speaker = winner of ≥ 3 of 5 bins
Otherwise no speaker for the bin
Only the PC compares badges
9 Output
Labeled CSV per sample
Speaker label per 500 ms bin
Input to downstream feature
extraction and engagement models
raw samples
A Observer reference
Independent observer panel
Per badge: Speaking / Quiet / Unknown
Detections hidden while recording
Intervals saved every 0.2 s
B Merge
Match by recording, badge, and
estimated acquisition time
Unknown time is excluded
Optional transition margin
C Evaluation on held-out whole recordings
1. Paper rule (steps 1 to 8, no training)
2. Shallow decision tree on paper features (ns, na, m)
3. Shallow decision tree on all 9 features (adds gyro and magnetic magnitude,
temperature, humidity, pressure, proximity); exported to C++ for step 4